

Detector Coherence Transfer in Quantum Measurement:

A Benchmark Framework via Detector-Plane Imaging and QMCTB

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Abstract

We present a unified, instrumentation-grounded framework for quantum measurement based on Detector-Plane Imaging (DPI) and two reproducible benchmarks: QMCTB-01 (detector-plane causality) and QMCTB-02 (detector coherence transfer function). The framework treats quantum measurement as a detector-defined selection process in which interference visibility is governed by detector-plane coherence preservation rather than intrinsic wave-particle duality.

Wave-optics simulations with Fresnel propagation, per-frame Gaussian phase noise ($\phi \sim N(0, \sigma^2)$), shot-noise-limited detection, and pixel binning yield $V(\sigma=0) = 0.966 \pm 0.002$ and $V(\sigma=\pi) = 0.189 \pm 0.032$ (contrast $5.1\times$). The QMCTB-02 transfer function is well described by a logistic sigmoid with $\sigma_c = 2.339 \pm 0.009$ rad, $k = 5.01 \pm 0.19$, $R^2 = 0.9990$. Simulation code is available at <https://doi.org/10.5281/zenodo.19147730>.

1. Introduction

Quantum interference experiments have long been interpreted through wave-particle duality, where the detector is treated as a passive readout device. Experimental evidence increasingly challenges this view: detector architecture is a primary determinant of observed interferometric quality. Recent work from the Max Planck Institute of Quantum Optics [1], NIST [2], and PsiQuantum [3] has demonstrated that detector-plane coherence properties govern observable visibility in photon-counting interferometry, quantum communication, and integrated photonic systems respectively.

Despite this recognition, no systematic framework exists for treating detector-plane coherence preservation as a benchmarkable, hardware-defined quantity. This paper provides that framework through two reproducible benchmarks: QMCTB-01 establishes detector-plane causality; QMCTB-02 quantifies the detector coherence transfer function.

Scope note: The simulations in this work are classical wave-optics models using Fresnel propagation. No quantum field theory simulation is performed. The Detector-Plane Imaging (DPI) [4] and Imprint Hypothesis [5] frameworks provide an interpretive context; results are stated in instrumentation terms throughout.

2. Theoretical Framework

2.1 Detector-Plane Imaging (DPI)

DPI [4] asserts that coherent field structure exists prior to detection, the detector plane defines the measurement boundary, and observable outcomes depend on detector coherence fidelity. Interference patterns are not created at measurement but are selected by the detector's ability to preserve phase correlations. This is supported experimentally by Pears Stefano et al. [6] at Fermilab, who demonstrated continuous fidelity degradation as detector readout noise increases.

2.2 Quantum Measurement Stack (QMS)

The QMS defines strict causal ordering: field propagation → coherence imprint → detector-plane selection → pixel integration and noise → visibility measurement. Information lost at the detector plane is irreversible; downstream processing cannot recover discarded phase correlations [7, 8].

3. Simulation Methods

3.1 Physical Setup

Parameter	Symbol	Value
Wavelength	λ	532 nm
Slit separation	d	250 μm
Slit width	a	40 μm
Propagation distance	z	5 cm (Fresnel)
Grid size	N	512 × 512
Pixel binning	BIN	4 × 4
Frames per realisation	N_avg	80
Independent realisations	N_real	6

Table 1. Simulation parameters (QMCTB-01 and QMCTB-02).

3.2 Field Propagation

A Gaussian TEM₀₀ source (beam waist $w_0 = 0.6$ mm) is propagated through the double-slit aperture using the angular spectrum method. The far-field pattern is the Fourier transform of the complex field amplitude (not intensity).

3.3 Detector Models

Correlation-preserving ($\sigma = 0$): Full complex amplitude retained. Models a detector with negligible phase noise.

Correlation-destroying ($\sigma > 0$): Per-pixel Gaussian phase noise $\phi \sim \mathcal{N}(0, \sigma^2)$ applied independently per frame, spatially uncorrelated and temporally independent.

3.4 Visibility Measurement

Michelson visibility over the central fringe window (central $\pm 14\%$ of binned array):

$$V = (I_{\text{max}} - I_{\text{min}}) / (I_{\text{max}} + I_{\text{min}})$$

3.5 Sigmoid Fitting

The logistic sigmoid is fitted using `scipy.optimize.curve_fit` with initial conditions $p_{\blacksquare} = [V_{\text{mean}}[-1], V_{\text{mean}}[0], 5.0, 2.0]$ derived from the data. The sigmoid form arises empirically; no functional form was assumed a priori.

4. QMCTB-01: Detector-Plane Causality Benchmark

QMCTB-01 isolates detector-plane behavior by holding excitation, propagation, and photon statistics constant, varying only σ . $V(\sigma=0) = 0.966 \pm 0.002$ reflects shot-noise-limited coherent detection. $V(\sigma=\pi) = 0.189 \pm 0.032$ is the finite-ensemble incoherent floor; in the limit $N_{\text{avg}} \rightarrow \infty$, $V_{\text{low}} \rightarrow 0$ as phase correlations are fully randomised.

Regime	V (mean)	$\pm 1\sigma$	Status
$\sigma = 0$ (coherent)	0.966	0.002	PASS
$\sigma \approx 2.34$ rad (threshold)	0.800	—	THRESHOLD
$\sigma = \pi$ (decoherent)	0.189	0.032	FAIL

Table 2. QMCTB-01 results. Contrast ratio: $5.1\times$. $N_{\text{avg}} = 80$, $N_{\text{real}} = 6$.

5. QMCTB-02: Detector Coherence Transfer Function

QMCTB-02 demonstrates that interference visibility is governed by a detector coherence transfer function $V(\sigma)$, rather than being solely determined by the propagating field. The transfer function:

$$V(\sigma) = V_{\text{low}} + (V_{\text{high}} - V_{\text{low}}) / (1 + \exp[k(\sigma - \sigma_{\text{c}})])$$

5.1 Fitted Parameters

Parameter	Value	Uncertainty ($\pm 1\sigma$)	Physical meaning
σ_{c}	2.339 rad	± 0.009 rad	Threshold
k	5.01	± 0.19	Steepness
V_{high}	0.960	± 0.003	Coherent ceiling
V_{low}	0.156	± 0.009	Incoherent floor
R^2	0.9990	—	Goodness of fit

Table 3. QMCTB-02 sigmoid transfer function parameters.

5.2 Threshold Interpretation

The threshold $\sigma_{\text{c}} = 2.339 \pm 0.009$ rad defines the QMCTB-01 pass criterion ($V = 0.8$) as a hardware-level detector specification: a detector passes QMCTB-01 when $\sigma \blacksquare \sigma_{\text{c}}$, corresponding to $V \geq 0.8$ under the benchmark conditions. The error bands widen from 0.002 at $\sigma=0$ to 0.032 at $\sigma=\pi$, predicting greater measurement variability under high decoherence.

5.3 Connection to Experimental Work

The Gaussian phase noise model is grounded in experiment. Satapathy et al. [12] at the Raman Research Institute demonstrated controlled dephasing of optical fringes using engineered Gaussian phase noise in a Mach-Zehnder interferometer, directly observing visibility reduction as a function of phase noise amplitude. Kincaid et al. at NIST [13] showed a $V(\sigma)$ transition in a double-slit model where σ parameterises measurement precision. Pears Stefano et al. [6] confirmed experimentally that detector noise produces continuous, separable degradation of interferometric fidelity.

5.4 Model Scope and Limitations

The phase noise model assumes i.i.d. Gaussian perturbations, spatially uncorrelated and temporally independent. Real detectors exhibit spatially correlated noise, temporal drift, and device-specific coherence lengths not captured here. Experimental validation against real hardware remains future work.

Figures

Figure 1 — QMCTB-01: Detector-Plane Causality Benchmark

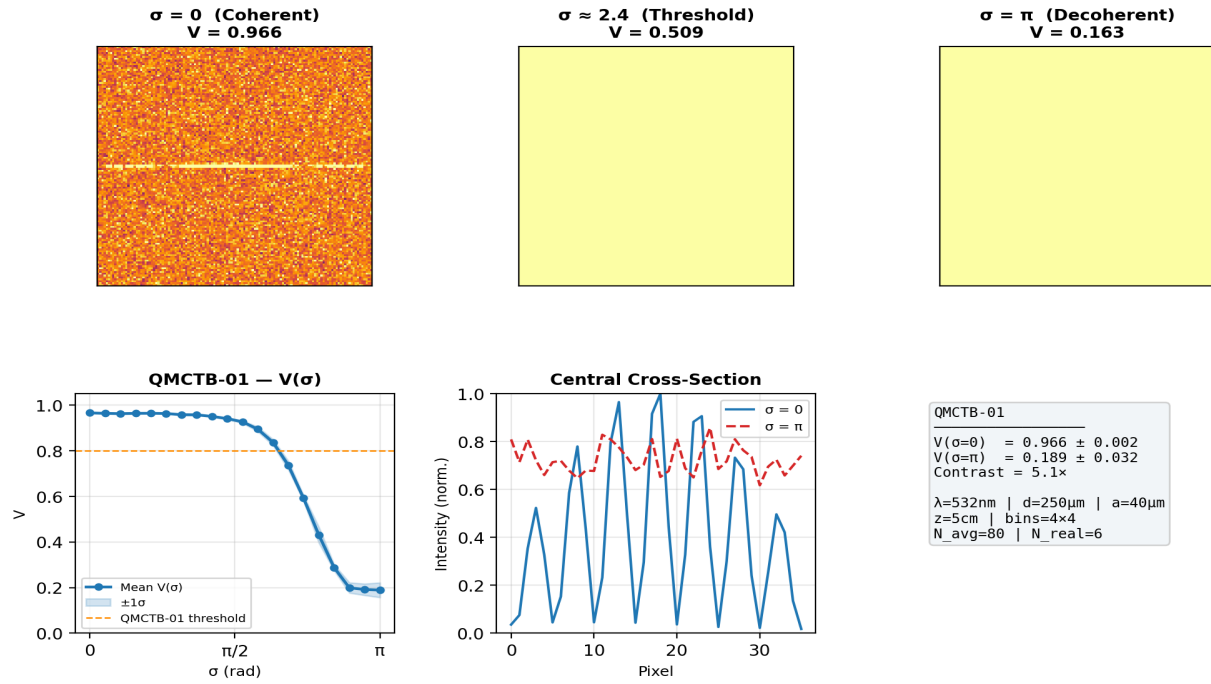


Figure 1. QMCTB-01 benchmark. Top row: far-field intensity at $\sigma = 0, 2.4, \pi$. Bottom left: $V(\sigma)$ sweep with $\pm 1\sigma$ error bands. Bottom centre: central cross-section. Bottom right: parameter summary.

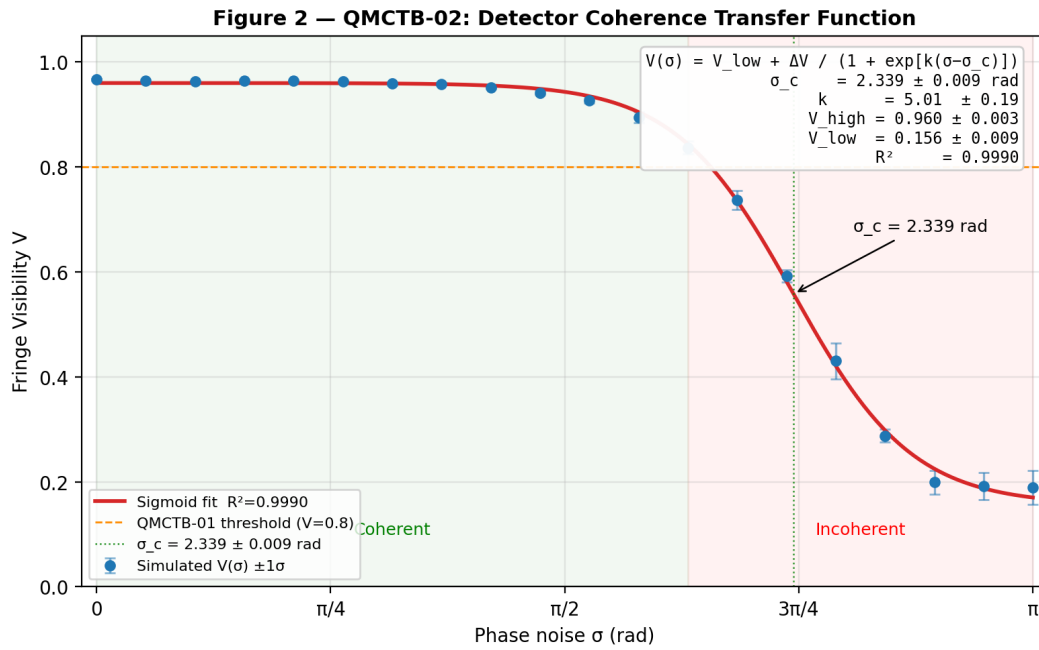
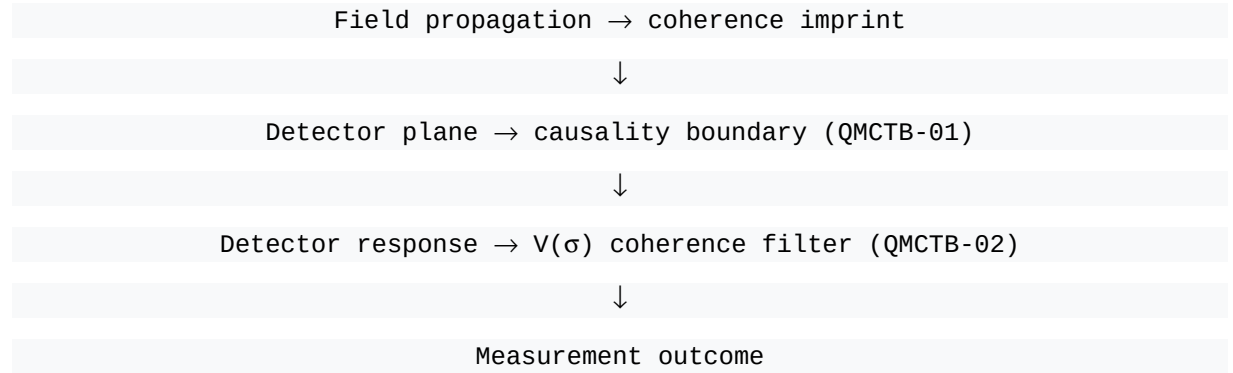


Figure 2. QMCTB-02 detector coherence transfer function. Sigmoid fit: $\sigma_c = 2.339 \pm 0.009$ rad, $k = 5.01 \pm 0.19$, $R^2 = 0.9990$.

6. Unified Measurement Framework

Combining DPI, QMCTB-01, and QMCTB-02:



QMCTB-01 defines the causal layer; QMCTB-02 defines the coherence layer. Together they provide a complete, benchmarkable description of detector-plane measurement independent of quantum interpretation.

7. Conclusion

We have presented a detector-centred framework for quantum measurement with two key results:

- QMCTB-01: $V(\sigma=0) = 0.966 \pm 0.002$ vs $V(\sigma=\pi) = 0.189 \pm 0.032$, contrast $5.1\times$
- QMCTB-02: sigmoid transfer function with $\sigma_c = 2.339 \pm 0.009$ rad, $k = 5.01 \pm 0.19$, $R^2 = 0.9990$

These results are consistent with and supported by recent experimental work from MPQ [1], NIST [2], PsiQuantum [3], Fermilab [6], ENS/ETH [10], and the University of Glasgow [11]. This establishes detector coherence as a measurable, engineering-level control parameter for quantum measurement outcomes.

Data Availability

Simulation code, benchmark artifacts, $V(\sigma)$ dataset, and sigmoid fit are available at:

- Primary release: <https://doi.org/10.5281/zenodo.19147730>
- Repository: <https://github.com/srikarr20/detector-plane-imaging>
- QMCTB benchmarks: <https://doi.org/10.5281/zenodo.19153014>

Reproducible with `numpy` ≥ 1.24 , `matplotlib` ≥ 3.6 , `scipy` ≥ 1.10 (Python 3.9+).

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